Property GIS Backend API Documentation

ASP.NET Core 8 - Feature Storage API

Base URL	http://10.79.222.188/backend/api/features
Content Type	application/json
Storage	JSON file - Data/features.json
Feature Identity	LayerId + FeatureId

Endpoints

Method	Endpoint	Purpose
GET	/api/features	Get all saved features
GET	/api/features/{layerId}/{featureId}	Get one saved feature
POST	/api/features	Validate and save a feature
DELETE	/api/features/{layerId}/{featureId}	Delete one saved feature

1. GET - Get All Features

Request: GET /api/features

Returns all features currently stored in the JSON feature store.

```
[
  {
    "layer": { "id": 0, "name": "ORC Properties" },
    "featureId": "TEST-1001",
    "geometryType": "polygon",
    "spatialReference": { "wkid": 2193 },
    "geometry": { "rings": [ ... ] },
    "attributes": { "OBJECTID": 1001 },
    "measurements": { "area": 10000, "perimeter": 400 },
    "savedAtUtc": "2026-10-25T07:00:00+00:00"
  }
]
```

Success: 200 OK

2. GET - Get One Feature

Request: GET /api/features/{layerId}/{featureId}

Finds a feature using the combination of LayerId and FeatureId.

Parameter	Type	Example	Description
layerId	integer	0	ArcGIS layer identifier
featureId	string	TEST-1001	Unique feature identifier inside the layer

Example: GET /api/features/0/TEST-1001

Success: 200 OK **Not found:** 404 Not Found

3. POST - Save Feature

Request: POST /api/features

Validates the request and stores the feature. Duplicate detection uses **LayerId + FeatureId**.

Request Body

```
{
  "layer": {
    "id": 0,
    "name": "ORC Properties"
  },
  "featureId": "TEST-1001",
  "geometryType": "polygon",
  "spatialReference": {
    "wkid": 2193
  },
  "geometry": {
    "rings": [
      [
        [1400000, 5000000],
        [1400100, 5000000],
        [1400100, 5000100],
        [1400000, 5000100],
        [1400000, 5000000]
      ]
    ]
  },
  "attributes": {
    "OBJECTID": 1001,
    "name": "Postman Test Property"
  },
  "measurements": {
    "area": 10000,
    "perimeter": 400
  }
}
```

Validation

Field	Rule
layer	Required
layer.name	Required and cannot be blank
featureId	Required and cannot be blank
geometryType	Required and must be supported
geometry	Required

Supported geometry types: **point, line, polyline, polygon, multipolygon**.

Responses

Status	Meaning
201 Created	Feature saved successfully
400 Bad Request	Request validation failed
409 Conflict	Feature with the same LayerId + FeatureId already exists
500 Internal Server Error	Unexpected server/storage error

Duplicate Response Example

```
{
  "message": "This feature has already been saved.",
  "layerId": 0,
  "featureId": "TEST-1001"
}
```

4. DELETE - Delete Feature

Request: DELETE /api/features/{layerId}/{featureId}

Deletes a stored feature using LayerId + FeatureId.

Example: DELETE /api/features/0/TEST-1001

Success Response - 200 OK

```
{
  "message": "Feature deleted successfully.",
  "layerId": 0,
  "featureId": "TEST-1001"
}
```

Not Found Response - 404

```
{
  "message": "Feature not found."
}
```

Storage Behavior

The API stores data in **Data/features.json**. The storage service uses a SemaphoreSlim to prevent overlapping read/write operations. When writing, the service first creates **features.json.tmp** and then replaces the main JSON file.

IIS Deployment Note

The IIS Application Pool identity must have Modify/Read/Write permission on the backend Data directory. Without write permission, POST and DELETE operations can return 500 Internal Server Error.

Postman Quick Test

Step	Method	URL
1	POST	http://10.79.222.188/backend/api/features
2	GET	http://10.79.222.188/backend/api/features
3	GET	http://10.79.222.188/backend/api/features/0/TEST-1001
4	DELETE	http://10.79.222.188/backend/api/features/0/TEST-1001
5	GET	http://10.79.222.188/backend/api/features/0/TEST-1001 (should return 404)

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